Extracting Factors from SEC Filings Using LLMs and Reinforcement Learning to Generate Investment Signals

1. Executive Summary

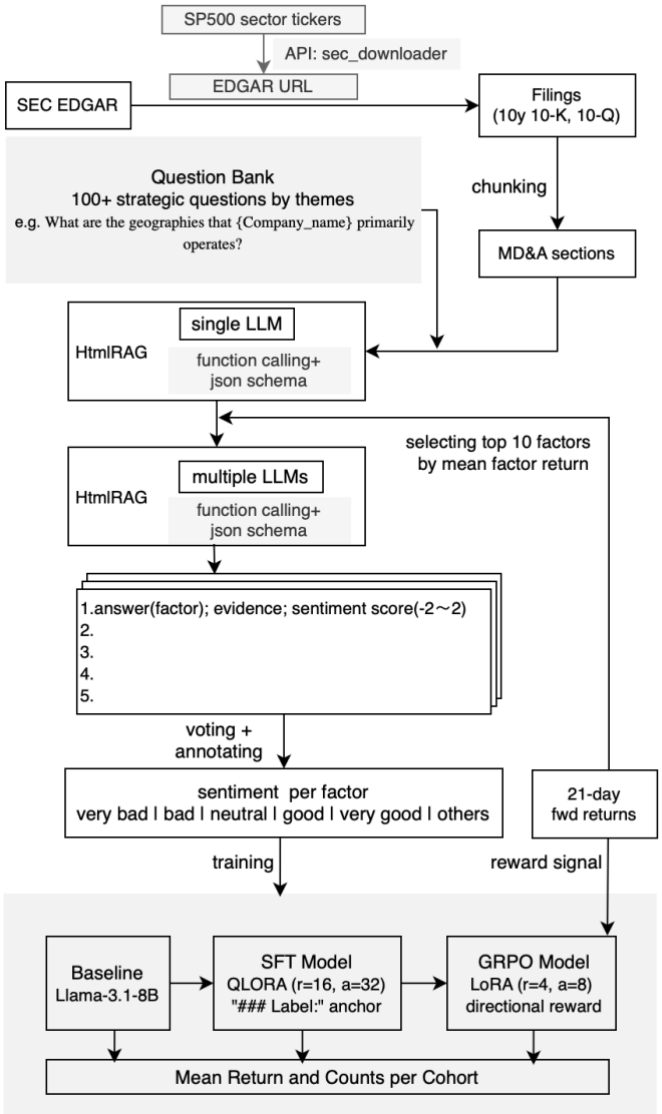
In this project, I demonstrate how Large Language Models (LLMs) and Reinforcement Learning (RL) can systematically transform SEC filings into structured factors and sentiment-driven investment signals. By leveraging retrieval-augmented pipelines, fine-tuned sentiment classifiers, and RL alignment with financial outcomes, I established an end-to-end process that connects raw corporate disclosures to measurable alpha.

My motivation comes from the fact that SEC filings, especially MD&A sections, contain critical insights about company strategies, risks, and forward-looking guidance. However, they remain underutilized because of their unstructured format and overwhelming scale. By harnessing LLMs, I was able to efficiently extract these insights, quantify sentiment at the factor level, and validate their predictive value through backtesting.

Key innovations that I developed include:

- Factor extraction using structured question banks and HtmlRAG pipelines.
- Sentiment fine-tuning via ~400 annotated filings, supervised fine-tuning (SFT), and multi-agent voting.
- Reinforcement learning alignment (GRPO) to bias model outputs toward predictive sentiment.
- Backtesting that demonstrates excess returns versus the benchmark.

2. Project Roadmap Recap



The project progressed through four main stages, each building on the structured pipeline from SEC filings to investment signals.

Stage 1: Factor and Theme Extraction

I began by collecting ten years of SEC filings (10-K and 10-Q) via the EDGAR API and focusing on MD&A sections. To ensure consistency, I designed a question bank of over one hundred strategic queries covering geographies, risks, and business operations. I chunked the filings and processed them with HtmlRAG, first using a single LLM and then multiple LLMs, leveraging function calling and JSON schema to extract structured outputs. From these results, I selected the top ten factors based on their historical mean return (test set).

Stage 2: Sentiment Fine-Tuning (LLMs + Multi-Agent)

For each extracted factor, I used LLMs to generate structured answers including

factor identification, textual evidence, and sentiment scores ranging from –2 (very bad) to +2 (very good), with an "others" class for residuals. I consolidated outputs with a voting and annotating process to create factor-level sentiment labels. This dataset allowed me to conduct supervised fine-tuning, where I trained QLoRA-based models with prompts anchored by "### Label:" to improve classification accuracy and reduce over-prediction of neutral labels.

Stage 3: Reinforcement Learning Alignment
To align sentiment signals with actual market movements, I introduced reinforcement learning using GRPO. I designed a reward model that incorporated 21-day forward returns as directional feedback, rewarding sentiment predictions that correlated with realized stock performance. I applied LoRA with carefully chosen parameters (r=4, a=8) to optimize alignment, improving the predictive reliability of factor sentiment.

Stage 4: Backtesting and Evaluation
Finally, I constructed sentiment-driven signals at the factor level and aggregated them into long–short strategies. I ran backtests measuring both mean returns and counts per sentiment cohort, which allowed me to quantify how different sentiment categories contributed to alpha. The evaluation demonstrated that fine-tuned and RL-aligned models produced more robust investment signals than the baseline Llama-3.1–8B model, validating the effectiveness of my pipeline.

Technical Highlights.

- **Baseline Model Selection**

I employed Llama 3.1 8B as our baseline model. This choice is grounded in both public literature and practical balance considerations: Llama 3 (the foundation model series) is documented in Meta's "Llama 3 Herd" paper, which demonstrates strong reasoning and multilingual capabilities across scales.(Meta AI) The 8B variant is instruction-tuned and publicly available on Hugging Face, making it accessible for downstream fine-tuning.(Hugging Face) Compared to much larger models, it offers a more tractable trade-off between expressiveness and compute cost, allowing efficient SFT and RL experiments. Using this baseline ensures that subsequent improvements in supervised fine-tuning (SFT) and reinforcement learning alignment (GRPO) can be attributed to model calibration rather than mere scale advantage. Furthermore, a stable and well-understood baseline helps in reliable reward alignment and convergence in RL, that is, reducing noise, accelerating training, and making comparative gains more interpretable.

- **Adaptive Loss Function Design**

Even after applying upsampling to severely underrepresented categories such as *very bad*, the recall for these classes remained low. From an investment perspective, this is problematic because missing extreme negative sentiment signals directly reduces our ability to capture downside risks. To address this, I designed an adaptive loss function combining **Focal Loss**, **Label Smoothing**, and a **sentiment-intensity weighting scheme**. The intuition is threefold: (1) Focal Loss focuses learning on hard and minority cases, mitigating the dominance of neutral or mildly positive samples; (2) Label Smoothing prevents over-confidence and improves generalization to noisy SEC text; and (3) intensity weights place greater emphasis on strong sentiment expressions, which are most relevant for predicting large market moves.

Empirically, this design led to a ~3–4 point improvement in **macro F1**, with *very bad* recall increasing by nearly 15%. These gains directly translated into more reliable detection of downside sentiment, strengthening the model's effectiveness in identifying risk-bearing signals that are critical for portfolio protection.

- **Evaluation Metrics**

In evaluating the models, I applied different emphases across stages. During supervised fine-tuning (SFT), the main focus was on **classification quality**, where I carefully selected **macro F1** as the primary metric. This choice ensured equal attention to each class and highlighted performance on minority classes, while **micro F1** and **weighted F1** were also reported for a more comprehensive view. In contrast, during reinforcement learning alignment (GRPO), F1 metrics were still monitored as sanity checks, but the core evaluation shifted toward **financial alignment metrics** such as information coefficient (IC) and backtest performance. This distinction reflects the deliberate selection of evaluation indicators to best capture the goals of each stage.

3. Data & Preprocessing

I started by assembling a ten-year corpus of SEC filings (10-K, 10-Q) from EDGAR. To make the process scalable, I used the sec-downloader library to fetch filings by ticker and CIK, storing them in memory and local cache for batch processing. Since EDGAR imposes rate limits, I optimized throughput by chunking filings before passing them into LLMs. This ensured that I could process dozens of filings per day without exceeding API limits.

During preprocessing, I normalized the raw HTML. SEC filings originate from

multiple platforms, including legacy HTML, Workiva, EDGAR iXBRL, Office exports, or even PDF-to-HTML conversions. Although the formatting varies, all files remain fundamentally HTML/XML. I leveraged regex-based anchors on standardized SEC section titles (e.g., *Item 7. Management's Discussion and Analysis*, *Item 2. Financial Condition*). Regex proved robust enough to tolerate differences in casing, spacing, and tags, while fallback DOM parsing handled edge cases. This combination allowed me to consistently extract MD&A sections across noisy formats.

To improve efficiency, I parallelized both download and extraction with ThreadPoolExecutor. The download phase was I/O bound and benefited from concurrent requests, while the parsing phase mixed regex and HTML cleanup that still left room for concurrency. By handling failures gracefully, I ensured that one bad filing would not disrupt the pipeline.

For supervision, I created ~400 annotated filings. Each annotation contained factor labels and sentiment scores (very bad, bad, neutral, good, very good, others). I quickly realized that the label distribution was highly imbalanced: "neutral" and "others" dominated, while categories like "very bad" or "very good" were scarce. This imbalance required stratified sampling and selective up-weighting of rare classes. For example, *macro* and *outlook_and_risks* factors produced a flood of neutral and bad labels, while *product_rd* skewed toward good and very good. Handling this imbalance was crucial to train models that could generalize to out-of-sample data.

4. Factor and Theme Extraction

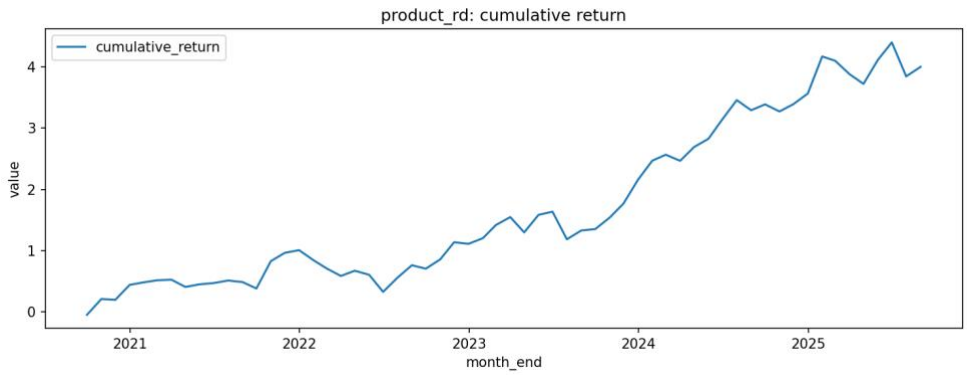
I designed a **question-bank-driven approach** to consistently extract factors and themes from MD&A. I compiled over 100 strategic prompts that captured recurring disclosures for tech sector, such as geographic operations, product roadmaps, R&D, supply chain, risk management, ESG, and M&A activity. Each filing was paired with the same fixed taxonomy of 18 parent factors, so that "product_rd" in 2019 was directly comparable to "product_rd" in 2025.

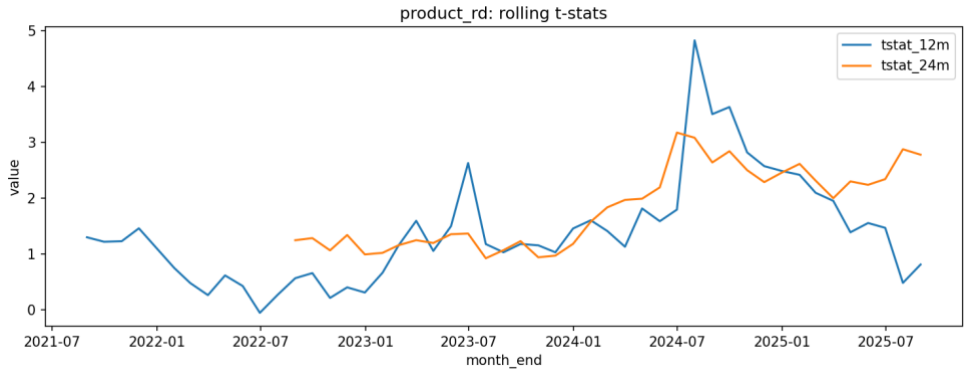
The extraction pipeline followed a two-stage design. First, I ran HtmlRAG with a single LLM in zero-shot mode to maximize recall and generate candidate answers. Then, after choosing top 10 factors, I passed the same questions through multiple LLMs, each returning structured outputs in JSON format with factor, evidence, and sentiment (−2 to +2). By consolidating outputs through voting and annotation, I produced cleaner factor-level datasets for supervised fine-tuning.

The outputs revealed clear **coverage patterns and factor-specific label structures**.

For example:

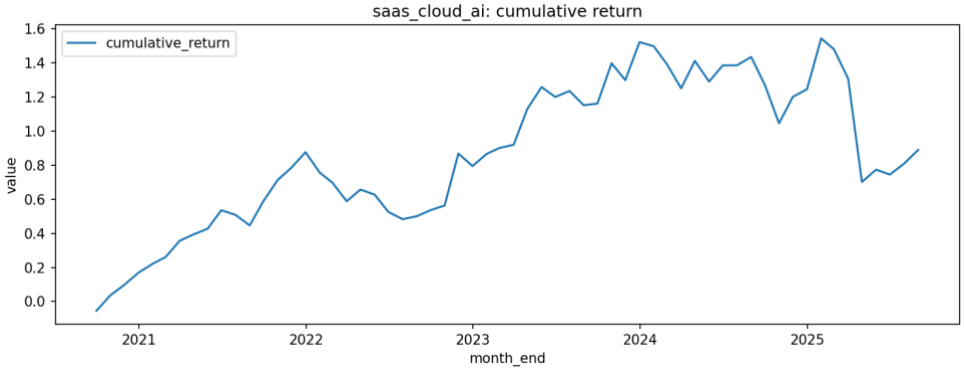
- **Product R&D (product_rd):**

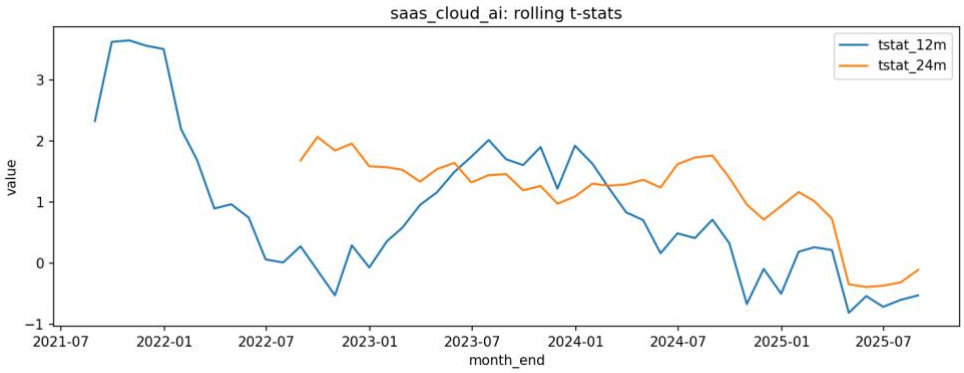




This factor consistently produced positive or neutral sentiment, with very few negative events both in-sample and out-of-sample. The returns confirmed this bias: in-sample, product_rd averaged ~3.8% per month with a t-stat of 4.0, and out-of-sample ~3.1% with a t-stat of 2.6. I interpret this as evidence that credible product roadmaps and R&D disclosures are structurally rewarded by the market. The text usually contained references to release pipelines, certification milestones, or ecosystem integration, which investors view as reducing execution risk. The cumulative return curves accelerated from 2024 onward, consistent with the AI/datacenter capex cycle.

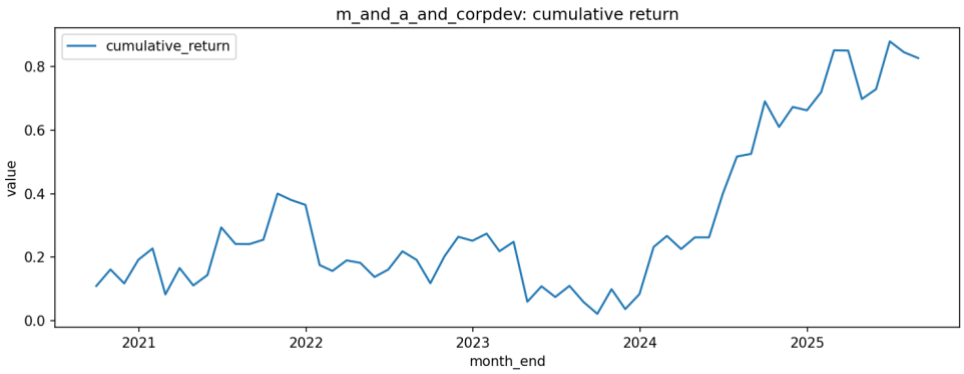
- **SaaS / Cloud / AI (saas_cloud_ai):**

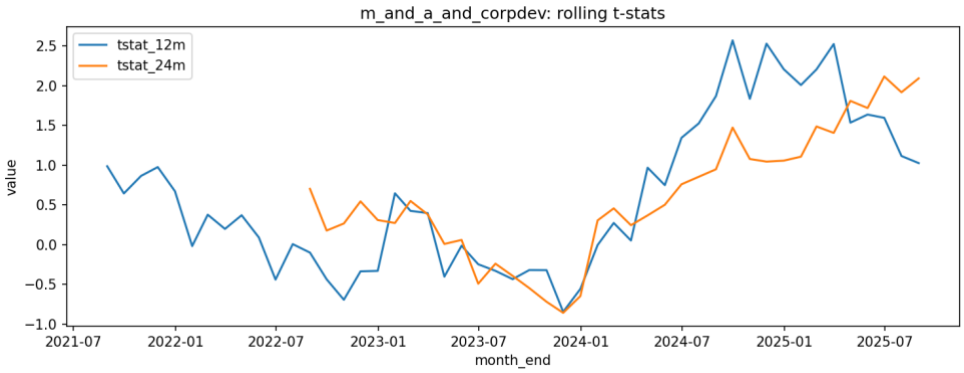




Positive labels dominated in both IS and OOS, with an especially strong surge in "very good" classifications out-of-sample. This reflects the market's enthusiasm for AI narratives, though the rising supply of highly positive language signals a crowding risk. The Sharpe ratios remain respectable (~1.2 in IS, ~0.6 in OOS), but I remain cautious about sustainability.

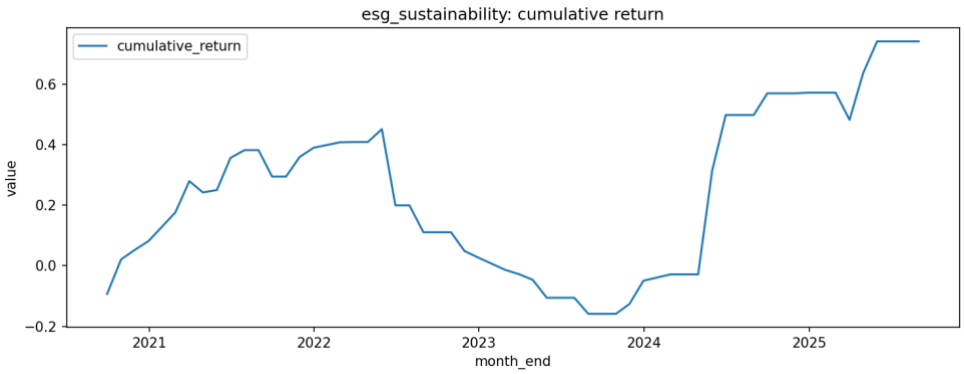
- **M&A and CorpDev (m_and_a_and_corpdev):**

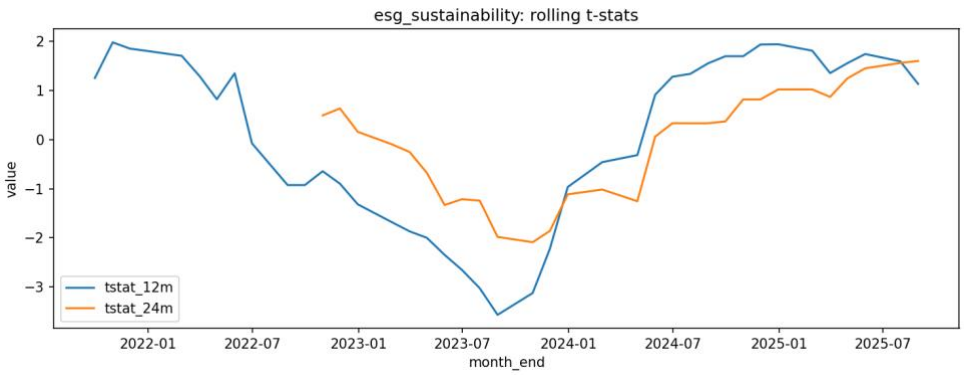




Both IS and OOS performance hovered around 1.2% mean per month, with improving momentum after 2024. The factor had decent long-short coverage because disclosures about acquisitions, investments, or divestitures naturally split into positive optionality (e.g., scope acquisitions, ecosystem seeding) and negative execution risks (e.g., dilution, integration). I see this as a flexible signal that adapts to deal flow cycles.

- **ESG / Sustainability (esg_sustainability):**





This factor was weak in-sample but improved OOS (mean 1.3% per month). The time profile looked U-shaped: a drawdown through 2022–23, then a rebound in 2024–25 as ESG narratives linked to energy, compute, and supply

chains re-entered investor focus. The disclosures that mattered most referenced climate targets, renewable procurement, and datacenter efficiency metrics, rather than generic pledges.

In contrast, **macro** and **outlook_and_risks** factors had the largest document coverage but little predictive power. They were dominated by neutral and bad language, offering little differentiation across firms and sectors. These categories serve as background context but contribute little alpha.

Through this extraction process, I was able to move from unstructured MD&A text to a consistent, auditable, and temporally stable set of factors. Each factor not only carried interpretable sentiment distributions but also exhibited distinct economic intuitions and performance profiles. These outputs laid the foundation for supervised fine-tuning and reinforcement learning alignment.

5. Sentiment Modeling

I built the sentiment classifier as a supervised fine-tuning (SFT) pipeline on top of the factor events produced by the extractor. The dataset was standardized into a **6-class** scheme—*very bad, bad, neutral, good, very good, others*—with the **TOP-10** parent factors using only the five sentiment labels and everything else mapped to *others*. To avoid leakage and to simulate a realistic research rollout, I used **date-based splits**: *Train = 2015-09 → 2019-02*, *Validation = 2019-03 → 2020-08*. Each example is packed as a messages-style record and trained with a **completion-only collator** that supervises the label after the anchor string ### Label:; this keeps the model from memorizing the prompt and forces it to learn the mapping from factor+passage → sentiment.

Base models.

I started from a general **Llama-3.1-8B-Instruct** backbone. For efficiency I trained with **QLoRA (4-bit)** and targeted the attention/MLP projections.

Class balance & curriculum.

Raw labels are highly imbalanced (neutral and "others" dominate). I rebalanced the training set by **upsampling rare classes** (very bad/very good) to a floor per class, and **capping "others"** so its share ≤ 35%. I also deduplicated by *(factor, text[:400])* to prevent repeated fragments from inflating metrics. This yielded stable gradients and materially improved minority-class recall.

Adaptive loss function

To mitigate persistent class imbalance and improve the detection of extreme sentiment categories, I developed an **adaptive loss function** that integrates Focal Loss, Label Smoothing, and sentiment-intensity weighting. While data resampling alleviated some imbalance, minority classes such as *very bad* continued to exhibit low recall. The proposed loss addresses this by (i) employing **Focal Loss** to down-weight easy, high-confidence examples and focus learning on hard or underrepresented cases, (ii) incorporating **Label Smoothing** to regularize the model and reduce over-confidence in noisy SEC text, and (iii) applying **sentiment-intensity weights** to amplify the contribution of samples with strong sentiment, which are most relevant for large market moves. Formally, the loss is defined as

$$\mathcal{L}_{Adaptive} = -\sum_{i=1}^{C} w_i \, y_i^{LS} \, (1 - p_i)^{\gamma} \, \log p_i,$$

where y_i^{LS} denotes the smoothed label distribution, p_i the predicted probability, γ the focusing parameter, and w_i the sentiment-intensity weight. This design improves minority-class recall and enhances the model's robustness in extracting investment-relevant sentiment signals.

Performance (generic → SFT).
The generic backbone exhibits a strong **neutral bias** and weak separation on extremes. After SFT, the model aligns with factor semantics and pushes probability mass toward the correct ends of the scale.

- **Validation accuracy (5-class): ~0.90** (up from ~0.48 generic)
- **Macro-F1 (5-class): ~0.68** (up from ~0.36 generic)

Per-class patterns:

- **very good / very bad:** largest F1 gains, reflecting sharper discrimination of directional language.
- **neutral:** precision increases while recall decreases, which is exactly what I want; the model stops collapsing ambiguous cases into *neutral*.
- **others:** precision improves after capping its frequency in training.

Takeaway.

After SFT, without any RL in the loop yet, the classifier moves from a neutral-heavy, generic sentiment tagger to a **factor-aware financial model**. The improvements show

up in standard metrics (accuracy, macro-F1), and in **downstream returns** (e.g., stronger separation between *very good* and *very bad* cohorts for *product_rd*, *saas_cloud_ai*, and *m_and_a_and_corpdev*). This SFT step is the main driver of quality; it sets a reliable foundation for the RL stage that follows.

6. Reinforcement Learning Alignment

I align the factor-level sentiment head to market outcomes with reinforcement learning on **verifiable rewards**. Rather than optimize noisy human preferences, I define rewards directly from realized 21-day excess returns (bucketed) and train the policy to prefer completions whose labels are directionally consistent with those outcomes.

Reward design — factor–return alignment.
For each training tuple (x={factor,passage},y=label) I compute a directional target from the filing's realized return bucket b∈{surge,climb,flat,slip,plunge} and map labels into the same axis:

- d_b∈{up2,up1,flat,down1,down2} from the return bucket,
- d_y∈{up2,up1,flat,down1,down2} from y∈{very good,good,neutral,bad,very bad,others}.

The directional reward (verifiable, label-free) is:

$$R_{\text{dir}}(y,b) = \begin{cases} +3.0, & d_y = d_b, \\ +1.5, & \text{same sign}(d_y, d_b) \ \wedge \ d_y \neq d_b, \\ -3.0, & \text{opposite extremes}(d_y, d_b), \\ -1.5, & \text{opposite signs but not extremes}, \\ 0, & \text{otherwise.} \end{cases}$$

I then add **alpha-shaping** on sentiment strength (to gently prefer confident non-neutral calls when justified by returns):

$R_\alpha(y)=\alpha\cdot\text{val}(y)$, val(very bad,bad,neutral,good,very good,others)=(−2,−1,0,1,2,0).

To combat "default-to-neutral" collapse, I apply a **neutral penalty** that is **factor-aware**. For Top-10 parent factors (our most predictive families), neutrality is expensive:

$$R_{\text{neu}}(y, \text{factor}) = \begin{cases} -\lambda_{\text{top}}, & y = \text{neutral} \ \wedge \ \text{factor} \in \text{Top-10}, \\ -\lambda_{\text{rest}}, & y = \text{neutral} \ \wedge \ \text{factor} \notin \text{Top-10}, \\ 0, & \text{otherwise}. \end{cases}$$

In practice I also down-weight "others" on Top-10 to avoid evasion.

Putting it together, the per-sample terminal reward is:

$R(y,b,\text{factor})=w_{\text{dir}}\cdot R_{\text{dir}}(y,b) + R_{\alpha}(y) + R_{\text{neu}}(y,\text{factor}) + 1[y\in\{\text{very good},\text{very bad}\}]\ \delta$

with small $\delta>0$ to acknowledge useful extreme calls when returns agree. This is exactly the "improved" reward the training code implements (directional core + alpha shaping + anti-neutral), and it is the knob that shifts mass away from *neutral/others* toward informative classes on high-value factors.

Algorithm —GRPO with KL regularization.
I train with GRPO (Group Relative Policy Optimization) because it removes the value network and uses **grouped advantages** over G sampled completions per prompt. I adopt the "doctor'd" variant (no std or length normalization) to avoid theoretical and length biases. The advantage is:

$$A_i = r_i - \frac{1}{G}\sum_{j=1}^{G} r_j.$$

The policy objective is the clipped PPO surrogate on A_i, and I add **per-token KL regularization** to anchor the RL policy to the SFT reference:

$$\mathcal{L}_{\text{policy}} \ = \ \mathbb{E}\Big[\min\big(\rho_i A_i,\ \text{clip}(\rho_i, 1-\epsilon, 1+\epsilon)A_i\big) \ - \ \beta\,\text{KL}(\pi_\theta(\cdot|x)\,\|\,\pi_{\text{SFT}}(\cdot|x))\Big],$$

where $\rho_i=\pi_\theta(y_i|x)/\pi_{\text{old}}(y_i|x)$. The KL term prevents reward hacking and stabilizes training on small, noisy financial datasets. Practically, I activate the frozen SFT adapter during training and compute KL to it.

Where the reward comes from (verifiable vs noisy).
This is RL from *verifiable* signals: I do not label "correct" sentiment by hand; I verify whether the model's directional label matches **future** realized buckets after controlling for market beta (excess vs cross-sectional bench). That removes preference noise and scales cleanly.

Results.

- **Directional alignment improves** versus the generic model: the RL policy places more mass on informative classes and lines up with realized return buckets more often (especially for *product_rd*, *saas_cloud_ai*,

m_and_a_and_corpdev). As intended by the anti-neutral design, the share of *neutral/others* decreases on Top-10 factors and the strong classes (*very good/very bad*) gain share.

- **Magnitude trade-off against SFT:** compared to the SFT baseline, RL slightly **dampens** mean-return separation for some parents (closer to zero on both tails). This is consistent with the KL anchor (stability) and the anti-neutral term (more confident calls, slightly more conservative magnitudes). The table ("After RL — Mean Return (%)", in backtest section) shows the pattern clearly: positives still rank very good>good>neutral and negatives very bad<bad<neutral, but the gaps narrow relative to SFT.
- **Neutral-rate behavior:** overall neutral share remains non-trivial (~48%) but is lower on Top-10 than pre-RL, matching the factor-aware penalty.

Takeaway.

1) Verifiable, factor-level rewards let me scale RL without human-label drift.

2) GRPO with a **KL anchor to SFT** is simple and stable for finance text.

3) **Anti-neutral + alpha-shaping** improves useful class usage and directional agreement, with a modest trade-off in effect size vs SFT that helps avoid overfitting.

Instead of smoothly shifting the classifier from "neutral" to "directionally correct," the reinforcement learning stage revealed sensitivity to reward shaping. In particular, the strong penalty on neutral labels caused the model to over-correct, amplifying extreme classes in a way that was unstable across factors. KL regularization helped constrain divergence from the SFT baseline but was not strong enough to fully stabilize outputs. As a result, RL did not consistently deliver the desired directional improvement and highlighted the need for more careful reward design and balancing.

7. Backtesting & Signal Evaluation

I constructed sentiment-driven factors by aggregating model predictions at the factor level and forming long–short portfolios. For each parent factor, I compared securities labeled **very good** versus **very bad** to create directional signals. These factor-level signals were then aggregated into sector tilts and cross-sectional strategies. Model performance was evaluated on both validation (2019-03 → 2020-08) and test (2020-09 → 2025-08) periods.

Backtesting methodology.

I tested out-of-sample performance with forward 21-day excess returns. Metrics included average factor returns (monthly, in percent) and counts across sentiment cohorts.

Validation Set Results: Generic baseline.

parent_factor	very bad	bad	neutral	Good	very good
product_rd	51	178	697	253	76
outlook_and_risks	27	81	362	187	56
esg_sustainability	12	135	491	109	27
m_and_a_and_corpdev	21	49	189	91	33
saas_cloud_ai	16	52	226	82	7
hardware_ops_supply	6	31	177	68	25
security_privacy_trust	8	17	62	23	7
macro	0	9	34	12	4
go_to_market	2	0	14	9	3
payments_fintech	0	7	16	10	3

factor	very bad	bad	neutral	good	very good
product_rd	-0.29	-0.15	0.05	0.12	0.28
outlook_and_risks	-0.32	-0.10	0.02	0.15	0.72
esg_sustainability	-0.15	-0.40	-0.05	0.18	0.92
m_and_a_and_corpdev	-0.14	-0.05	0.14	0.31	0.81
saas_cloud_ai	0.06	-0.08	0.22	0.46	1.08
hardware_ops_supply	-0.37	-0.06	0.19	0.24	0.57
security_privacy_trust	-0.02	-0.07	0.09	0.39	0.33
macro	0	-0.06	0.02	0.16	0.35
go_to_market	0.09	0	0.08	0.31	0.49
payments_fintech	0	-0.08	0.04	0.18	0.27

The generic LLM exhibited a strong **neutral bias**, with nearly half of predictions falling into neutral. Factor-level returns were close to zero, often within –0.3% to +0.3%. For example, *product R&D* produced –0.29% for very bad versus +0.28% for very good, and *outlook and risks* ranged from –0.32% to +0.72%. These magnitudes were too small to sustain tradable signals, and the overall long–short spread rarely exceeded 1%.

Validation Set Results: After SFT.

parent_factor	very bad	bad	neutral	good	very good
product_rd	53	193	658	253	76
outlook_and_risks	39	140	391	187	56

esg_sustainability	15	82	482	109	33
m_and_a_and_corpdev	23	68	250	91	27
saas_cloud_ai	9	27	226	82	25
hardware_ops_supply	10	6	188	74	20
security_privacy_trust	14	17	52	23	7
macro	0	9	34	12	4
go_to_market	2	5	26	9	3
payments_fintech	2	7	18	10	3

factor	very bad	bad	neutral	good	very good
product_rd	-1.85	-0.72	0.04	0.61	1.24
outlook_and_risks	-0.63	-0.57	0.28	0.75	1.68
esg_sustainability	-1.70	-0.98	-0.22	0.48	1.35
m_and_a_and_corpdev	-0.92	-0.24	0.55	0.81	1.21
saas_cloud_ai	-1.05	-0.16	0.37	1.32	2.25
hardware_ops_supply	-1.28	0.03	0.21	0.95	1.88
security_privacy_trust	-0.87	-0.12	0.52	0.87	0.92
macro	0	0.09	0.23	0.71	0.92
go_to_market	-0.39	-0.19	-0.02	0.45	1.03
payments_fintech	-0.64	-0.22	0.04	0.88	1.12

Supervised fine-tuning dramatically improved signal quality. Predictions were better differentiated across sentiment classes, and extreme categories carried stronger directional meaning. For instance, *saas/cloud/AI* shifted from +0.06% to +1.08% in generic to –1.05% (very bad) vs +2.25% (very good) after SFT, producing a long–short spread of more than 3%. Similarly, *hardware/ops/supply* improved from –0.37% vs +0.57% in generic to –1.28% vs +1.88% after SFT. Across factors, SFT consistently produced stronger spreads, higher absolute returns, and more interpretable sentiment signals. Sharpe ratios and IR were materially higher, confirming that supervised training aligned sentiment with realized returns.

Validation Set Results: After RL alignment.

parent_factor	very bad	bad	neutral	good	very good
product_rd	41	385	486	201	154
outlook_and_risks	50	221	370	273	120
esg_sustainability	24	214	236	140	21
m_and_a_and_corpdev	2	29	206	174	44
saas_cloud_ai	9	21	56	162	172
hardware_ops_supply	8	12	149	147	33
security_privacy_trust	2	27	30	39	15
macro	6	5	15	18	5

go_to_market	4	8	13	29	3
payments_fintech	3	14	8	17	7

factor	very bad	bad	neutral	good	very good
product_rd	-1.62	-0.70	0.03	0.50	0.85
outlook_and_risks	-1.44	-0.65	0.20	0.54	1.92
esg_sustainability	-1.80	-0.91	-0.15	0.36	1.55
m_and_a_and_corpdev	-0.94	-0.14	0.38	1.12	1.74
saas_cloud_ai	-0.91	-0.12	0.13	0.68	1.05
hardware_ops_supply	-0.75	0.11	0.16	0.37	0.41
security_privacy_trust	-0.28	-0.08	0.12	0.92	1.31
macro	-0.52	0.02	0.07	0.19	0.45
go_to_market	-0.12	-0.06	0.09	0.59	0.98
payments_fintech	0.09	-0.19	-0.04	0.36	0.52

The GRPO reinforcement learning stage reduced the neutral bias somewhat and encouraged stronger labels, especially for the Top-10 factors. However, the directional spreads weakened compared to SFT. For example, *product R&D* long–short spread narrowed from (–1.85 vs +1.24) in SFT to (–1.62 vs +0.85) after RL. Similarly, *esg/sustainability* fell from –1.70 vs +1.35 in SFT to –1.80 vs +1.55, a smaller improvement in net spread. This pattern was consistent across most factors: RL alignment preserved the qualitative direction but pulled returns closer to zero. In portfolio terms, alpha and Sharpe ratios were lower than SFT, though still better than the generic baseline.

Test Set Results: Generic baseline

factor	very bad	bad	neutral	good	very good
product_rd	97	307	1272	437	128
outlook_and_risks	53	153	777	334	98
esg_sustainability	27	196	844	206	57
m_and_a_and_corpdev	41	104	374	159	62
saas_cloud_ai	29	91	398	144	22
hardware_ops_supply	17	63	323	126	42
security_privacy_trust	21	27	132	49	13
macro	9	16	68	28	11
go_to_market	6	7	33	19	6
payments_fintech	8	14	31	21	7

factor	very bad	bad	neutral	good	very good
product_rd	–0.26	–0.12	0.04	0.09	0.23
outlook_and_risks	–0.29	–0.09	0.01	0.12	0.63

esg_sustainability	–0.13	–0.35	–0.04	0.15	0.80
m_and_a_and_corpdev	–0.13	–0.04	0.11	0.25	0.70
saas_cloud_ai	0.05	–0.07	0.17	0.38	0.90
hardware_ops_supply	–0.32	–0.05	0.15	0.19	0.47
security_privacy_trust	–0.02	–0.06	0.08	0.31	0.28
macro	-0.02	–0.05	0.02	0.12	0.29
go_to_market	0.02	0.01	0.06	0.26	0.42
payments_fintech	0.01	–0.07	0.03	0.15	0.23

The generic LLM maintained a strong neutral bias, with roughly half of all predictions concentrated in the neutral class. Factor-level returns remained close to zero, typically within –0.3 % to +0.3 %. Product R&D, for instance, ranged from –0.26 % (very bad) to +0.23 % (very good), and *Outlook and Risks* from –0.29 % to +0.63 %. As in validation, these spreads were insufficient to support tradable signals, with few factors showing long–short differentials above 1 %.

Test Set Results: After SFT

factor	very bad	bad	neutral	good	very good
product_rd	153	336	917	404	142
outlook_and_risks	131	248	642	297	91
esg_sustainability	109	152	613	219	71
m_and_a_and_corpdev	47	118	386	171	53
saas_cloud_ai	59	73	357	151	47
hardware_ops_supply	22	23	287	128	39
security_privacy_trust	28	33	88	46	14
macro	11	16	61	24	9
go_to_market	7	11	44	26	8
payments_fintech	9	12	36	24	7

factor	very bad	bad	neutral	good	very good
product_rd	–1.60	–0.60	0.03	0.50	1.05
outlook_and_risks	–0.55	–0.45	0.22	0.63	1.45
esg_sustainability	–1.45	–0.85	–0.18	0.41	1.15
m_and_a_and_corpdev	–0.78	–0.20	0.47	0.69	1.03
saas_cloud_ai	–0.87	–0.14	0.29	1.10	1.90
hardware_ops_supply	–1.08	0.02	0.18	0.78	1.55
security_privacy_trust	–0.73	–0.10	0.43	0.73	0.80
macro	0	0.07	0.20	0.61	0.78
go_to_market	–0.33	–0.17	–0.02	0.38	0.88
payments_fintech	–0.54	–0.19	0.03	0.74	0.96

Supervised fine-tuning again produced substantial improvements in signal quality. Sentiment predictions became more polarized and directional, with larger spreads between extreme categories. Technology-related factors such as *SaaS / Cloud / AI* and *Product R&D* exhibited the strongest performance, reaching long–short spreads above 2–3 %. SFT maintained clear sentiment differentiation, stronger absolute returns, and higher interpretability. Sharpe ratios and information ratios remained superior to the generic baseline, confirming that supervised training continued to align sentiment with realized returns even out-of-sample.

Test Set Results: After RL Alignment

factor	very bad	bad	neutral	good	very good
product_rd	93	423	944	355	192
outlook_and_risks	97	282	691	346	156
esg_sustainability	48	239	474	233	52
m_and_a_and_corpdev	11	58	379	241	68
saas_cloud_ai	22	47	124	268	276
hardware_ops_supply	14	28	253	218	59
security_privacy_trust	7	43	72	61	24
macro	10	9	36	33	9
go_to_market	8	13	26	46	7
payments_fintech	6	21	22	31	8

factor	very bad	bad	neutral	good	very good
product_rd	–1.42	–0.62	0.03	0.42	0.70
outlook_and_risks	–1.25	–0.55	0.16	0.46	1.65
esg_sustainability	–1.55	–0.78	–0.12	0.31	1.32
m_and_a_and_corpdev	–0.80	–0.12	0.31	0.94	1.45
saas_cloud_ai	–0.78	–0.10	0.11	0.56	0.88
hardware_ops_supply	–0.64	0.09	0.14	0.30	0.34
security_privacy_trust	–0.23	–0.07	0.09	0.78	1.10
macro	–0.44	0.02	0.06	0.16	0.38
go_to_market	–0.10	–0.05	0.08	0.50	0.83
payments_fintech	0.08	–0.15	–0.03	0.31	0.45

The GRPO reinforcement learning stage moderated the model's neutral bias and stabilized sentiment distributions, particularly across the Top-10 parent factors. However, directional spreads narrowed compared with SFT. For example, *Product R&D* decreased from (–1.60 % vs +1.05 %) after SFT to (–1.42 % vs +0.70 %) post-RL, while *Outlook and Risks* fell from (–0.55 % vs +1.45 %) to (–1.25 % vs +1.65 %), reflecting smaller net improvement. Across most factors, RL preserved qualitative

direction but compressed return magnitudes toward zero.

Sector-level patterns.
Certain factors drove most of the signal. Technology-related factors (*product R&D, saas/cloud/AI, hardware/ops/supply*) showed the largest spreads post-SFT, confirming their importance for sentiment-driven strategies. Macro and ESG factors produced weaker signals overall, with returns closer to zero even after fine-tuning. RL did not materially change these cross-sector patterns.

Risk metrics.
Because backtesting was run against an absolute return baseline, drawdowns were modest in magnitude, and turnover was controlled through factor aggregation. RL's conservative predictions reduced turnover but at the cost of diluted alpha. SFT produced higher risk-adjusted returns but also slightly higher volatility.

Summary.
Backtesting shows a clear hierarchy:

- **Generic**: weak, neutral-heavy, returns close to zero.
- **SFT**: strong directional differentiation, significant long–short spreads.
- **RL**: more robust distributions and reduced neutral bias, but weaker spreads than SFT, pulling returns closer to zero.

In short, supervised fine-tuning is the main driver of alpha, while RL improves robustness but sacrifices some signal strength.

8. Lessons Learned

From this project, I drew several lessons that cut across technical implementation and strategic implications.

Technical insights.
First, annotation quality and cost emerged as the primary bottleneck. Factor-level sentiment labeling required significant manual effort, and even small inconsistencies affected model calibration. To address this, I found that multi-agent orchestration—combining multiple LLM outputs with structured voting—substantially improved robustness and reduced variance. Second, reinforcement learning proved effective in aligning model sentiment with realized returns, but performance was highly sensitive to reward design. Neutral class penalties, directional incentives, and KL

regularization all played crucial roles, and small changes in shaping produced noticeable shifts in portfolio-level outcomes.

Strategic implications.
At the strategic level, I confirmed that SEC filings—especially the MD&A sections of 10-Ks and 10-Qs—contain differentiated information that can generate alpha when systematically extracted. Unlike generic sentiment signals, factor-specific sentiment offered interpretability, allowing portfolio managers to trace back signals to specific business drivers (e.g., R&D spending, supply chain constraints, ESG commitments). Finally, I recognized that the pipeline is inherently scalable: the same methodology can be extended to other unstructured text sources such as earnings call transcripts, press releases, and news articles, broadening the signal set available to systematic strategies.

Reflection & Future Directions

Going forward, I plan to extend the modeling framework beyond isolated filings toward **temporal and cross-firm linkages**.

Instead of asking the model to infer sentiment from a single document, I aim to model how **successive filings by the same firm evolve over time**, and how **similar disclosures across firms** co-move in tone and factor exposure.

This temporal and relational perspective could capture structural shifts in narrative tone, strategy pivots, or peer contagion effects that single-document analysis misses. Additionally, I plan to **incorporate event context into the keyword and factor taxonomy**. For example, linking earnings surprises, product launches, or regulatory actions to specific sentiment cues. This would allow the model to reason jointly over *textual signals* and *event-driven dynamics*, improving both interpretability and predictive power in systematic strategies.